

EDUCATION

BSc Computer Science, Carnegie Mellon University

August 2021 - Spring 2025

- Algorithm Design and Analysis Teaching Assistant (January 2024 – January 2025)
- Relevant Coursework: Compiler Design, Parallel and GPU Computing, Machine Learning, Foundations of Programming Languages, Computer Graphics

WORK EXPERIENCE

Tutor Intelligence

Boston

Member of Technical Staff, Full-time

August 2025 - present

- Designed and implemented model rollout infrastructure for autonomous robot control with VLA models
- Developed ownership over real-time long-distance teleoperation control system for the humanoid data factory, DF1 (<https://tutorintelligence.com/blog/building-a-100-robot-data-factory-toward-factory-ready-ai>)
- Created Tutor’s model rollout observability platform, used data replay to find VLA bugs, implemented real time action chunking for real-time model rollouts, finetuned models
- Optimized video and point-cloud streaming pipeline by >8x, achieving a real-time 3D VR environment for Tutor’s in-house robotic teleoperation system

PEAK6 Capital Management

Chicago

Quantitative Research Intern

June 2024 - August 2024

- Designed and integrated market data pipeline observability using OpenTelemetry and Jaeger code instrumentation. Explored optimization with Cython.
- Trained and quantitatively compared deep RNN, LSTM, and factorization model architectures for effectiveness in embedding time-series data using company stock market data and TensorFlow

Boston Dynamics AI Institute

Boston

R&D Systems Intern

June 2023 - August 2023

- Created an asynchronous Python scheduler to simultaneously direct four Spot robots to perform a sequence of interactive tasks over a shared environment with an 89% completion rate under high-stress, dynamic demo runs
- Designed and developed mapping system for mobile robot rearrangement of environment under human-guidance with high-level planning to correct for object misplacements

IBM

Boston

Security Systems Development Intern

May 2022 - August 2022

- Reimplemented authentication request process for backend refactor of major API security software product in C++
- Developed a configuration parser and API mocking system using Google Mock to enable efficient and flexible end-to-end testing leveraged by team over entire effort

TECHNOLOGY SKILLS

Languages: C++, Python, C, OCaml, Java

Development tools: MongoDB, MySQL, Node.js, PHP, HTML/CSS, OpenCV, OpenTelemetry, Google Test/Mock, Vulkan

System tools: ROS1, ROS2, Linux, Docker, CMake, CUDA, OpenMP, MPI

Data tools: PyTorch, TensorFlow, MATLAB, Pytorch3D

NOTABLE PROJECTS

- Partner-developed and optimized an SSA-based C compiler in OCaml with >90% execution speed of -O1 GCC and a novel automatic copy-collected garbage collection memory system (January 2024 - May 2024)
- Augmented open-source SLAM system (ORBSLAM) with robustness to dynamic points, reducing map reconstruction error by 9.4% over TUM dataset videos where the system did not previously diverge and preventing map divergence for largely dynamic scenes

RESEARCH

Drone Search and Rescue Effort, AirLab, CMU

Research Assistant

August 2024 – December 2024

- Explored dense, language-aligned 3D semantic scene representations for open-vocabulary navigation queries
- Found SOTA results for automated semantic pixel feature labelling using SAM2 to modify CLIP’s attention mechanism

Autonomous Underwater Vehicle Team, CMU

Robotic Perception Lead and Co-Founder

August 2021 - May 2024

- Placed 3rd place overall in 2022 and as finalists every year in international RoboSub competition
- Developed global mapping and tracking system for underwater object awareness using EKF SLAM to globally interpolate relative object positions from a fusion of perception sensors
- Wrote end-to-end perception stack, developing YOLO-based object detection system, implementing drivers for camera and sonar system, writing signal processing for noise reduction of sonar echoes